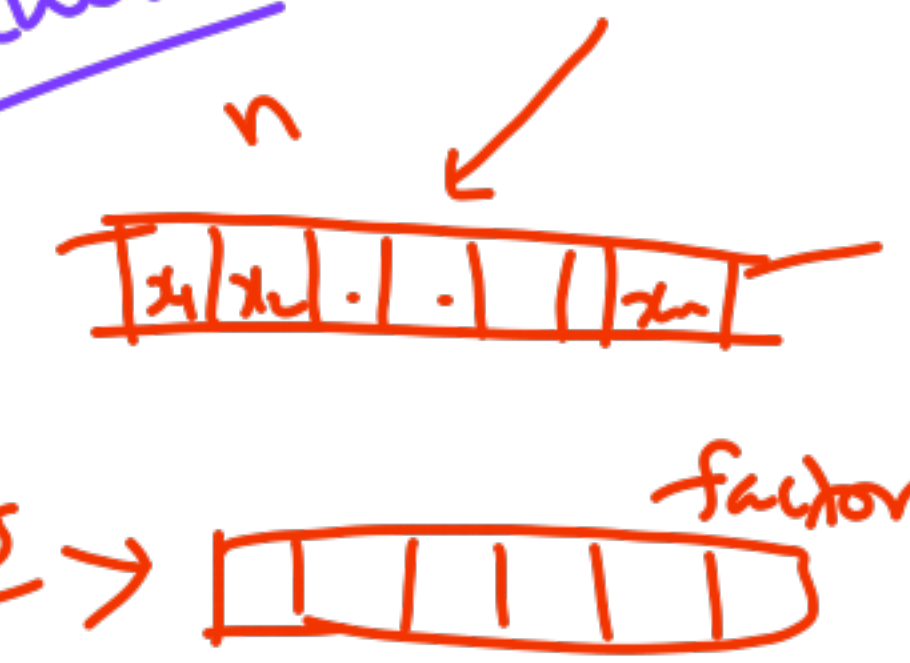


31.10.25

NP-Completeness

Polynomial # steps —
Turing Machine



Non-deterministic

choice

Verify in polynomial time

Verifier

"given by Oracle"

"Guess"

one correct choice $\Rightarrow$ accept

Search

don't know

find a solution

- Quiz 3
- Lectures
 - 1. 3/10
 - ~~2. 5/11~~
 - 3. 7/11
- (Assignment - 6)
- Final Exam (15/11)

Polynomial Time

Deterministic

DTM

Count # of steps

Input Size

graph

$n_1 \# n_2 \# n_3 \# n_4$
edge 1 edge 2

Knapsack

x unary

numbers

log

binary

Standard
↓ site
3 lectures

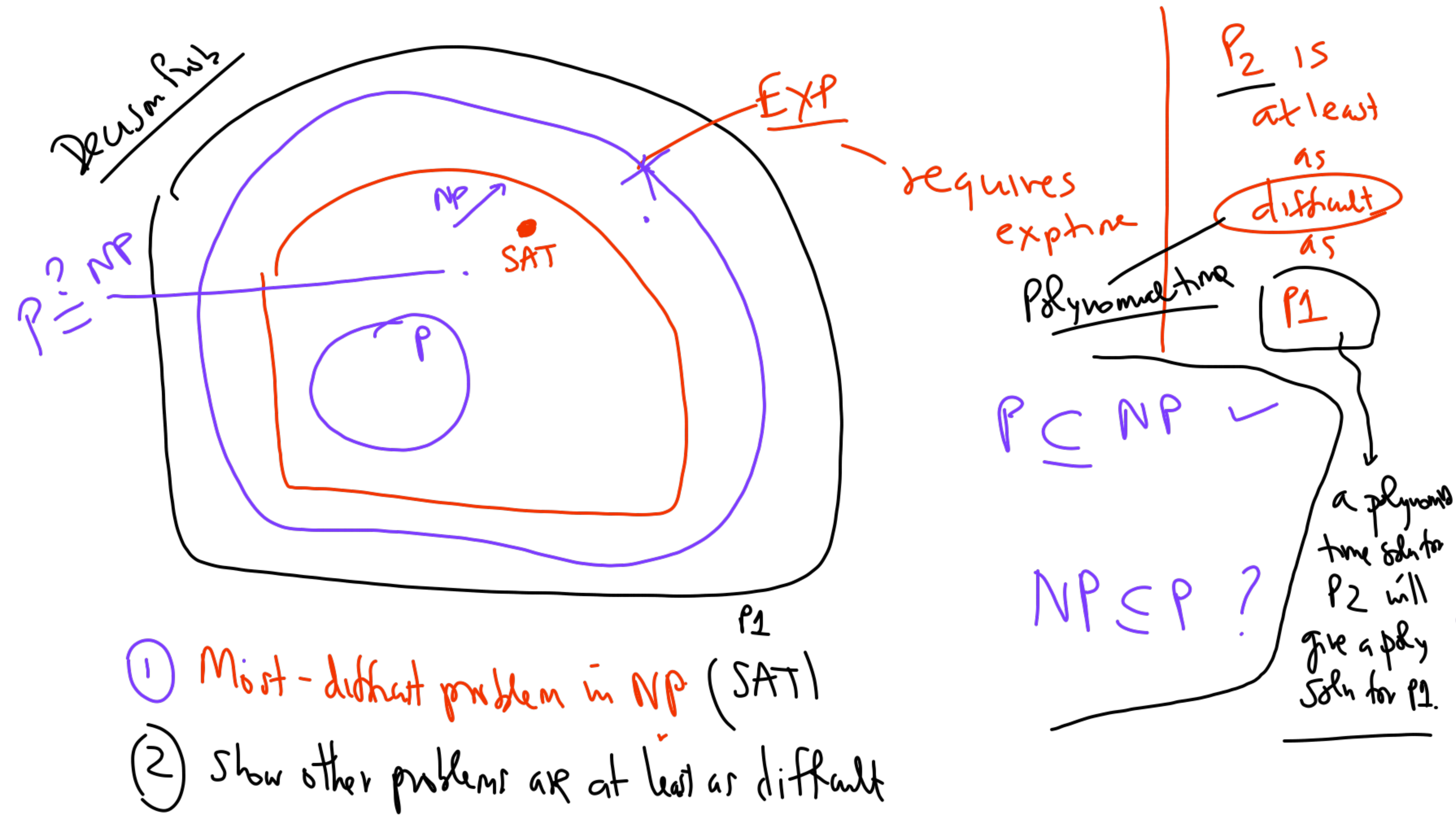
Decision Problems

vs
Computations
or
Optimizations

Polynomial time
algorithms

$ID_0 \rightarrow ID_1 \rightarrow ID_2$

$\boxed{1, q_1, x_2, \dots}$



Show that 3-SAT is at least as difficult as SAT ✓

Any prop logic formula is a CNF formula

Literal	$P, \neg P$
Clause	$P \vee Q \vee \neg R \vee \neg S \dots$
CNF	$C_1 \wedge C_2 \wedge \dots \wedge C_k$

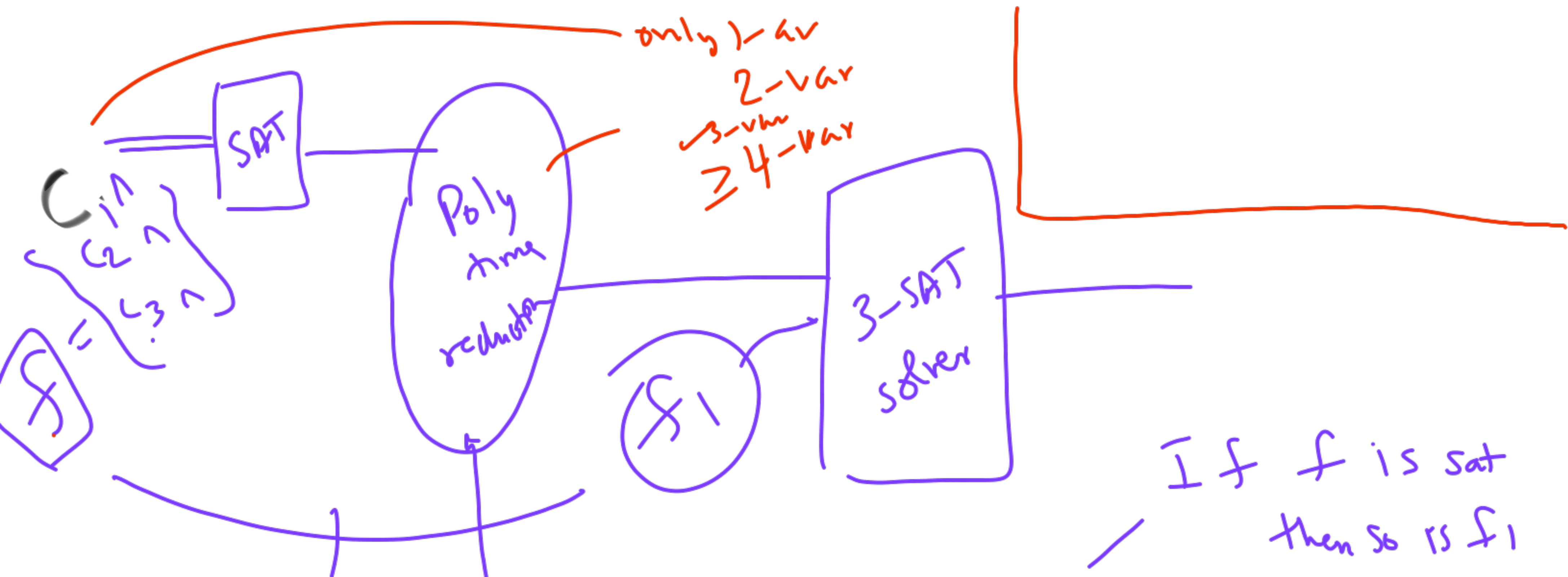
C_1	$x_1 \vee \bar{x}_2 \vee x_4$
C_2	$\bar{x}_3 \vee x_5 \vee x_1 \vee \bar{x}_7 \vee \dots$
$\vdots$	
C_k	

every clause exactly 3-literals

is false

Vars	truth
x_1	T
x_2	F
x_3	F
$\vdots$	
$\vdots$	

model



If f is sat
then so is f_1

Convert f to f_1 → Equi Satisfiable
(SAT) (3-SAT)

f
 $a \vee b$

Equivalent f_1 ~~not same~~
 $(a \vee c) \wedge (\neg c \vee b)$ $\left\{ \begin{array}{l} a, c = f \\ b = \text{true} \end{array} \right.$

If f_1 is sat
then so is f

SAT $\longrightarrow$ 3-SAT

$\cancel{x_{\text{new vars}}}$

$(l_1 \vee x_1 \vee x_2)$
 $(l_1 \vee \bar{x}_1 \vee x_2)$
 $(l_1 \vee x_1 \vee \bar{x}_2)$
 $(l_1 \vee \bar{x}_1 \vee \bar{x}_2)$

$x = l_1$
 $\bar{y} =$

$l_1 \vee l_2 \rightsquigarrow (l_1 \vee l_2 \vee x) \wedge$
 $(l_1 \vee l_2 \vee \bar{x})$

new variable

$\rightarrow l_1 \vee l_2 \vee \dots \vee l_{100}$

97 new vars

$x_1 \dots x_{97}$

$(l_1 \vee l_2 \vee x_1)$

$(\bar{x}_1 \vee l_3 \vee x_2)$

$(\bar{x}_2 \vee l_4 \vee x_3)$

$\dots$

$(\bar{x}_{97} \vee l_{99} \vee l_{100})$

98 clause

Recap

any { NTM
polynomial
verifier. } ✓

Stanford
lecture slides

H/W
2/11

SAT

is the
most
difficult
(NP-complete)

3-SAT ✓

a

Subset-sum

Knapsack

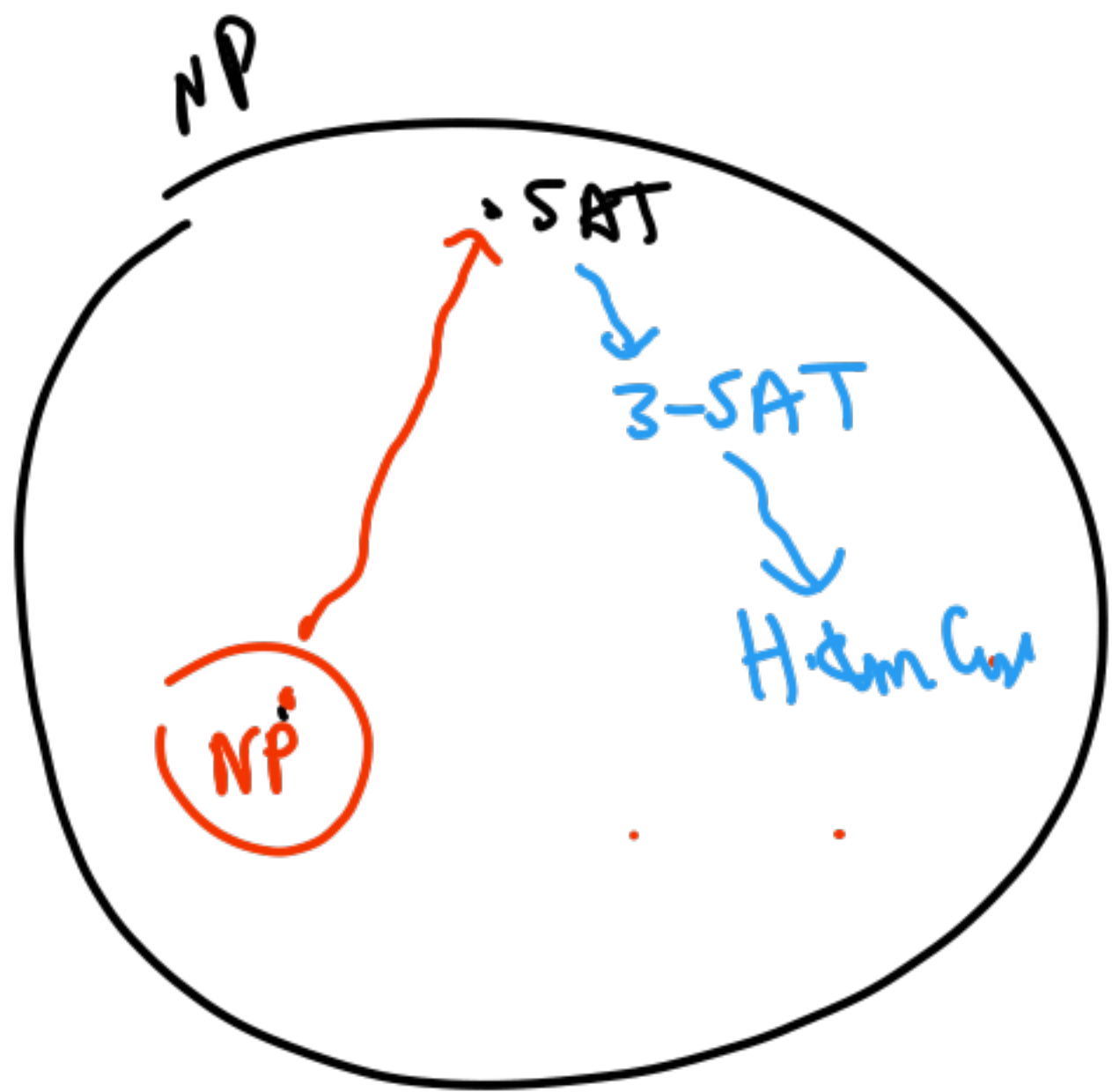
b

Hamiltonian
Circuit

(TSP)

clique

Vertex-cover



Yes ✓ 3SAT →

known NP-complete

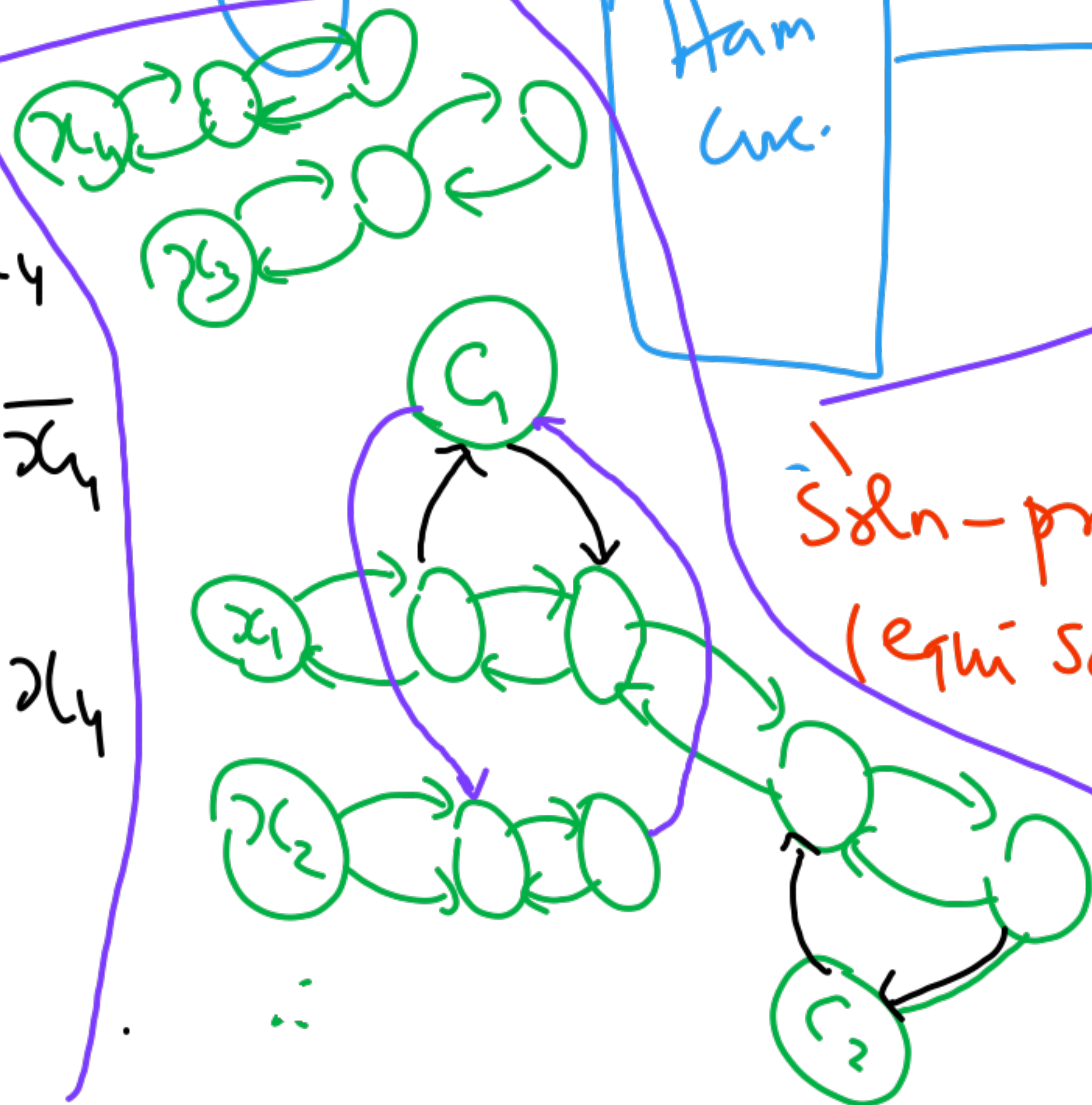
Polynomially (reducible)

Ham. Circ.

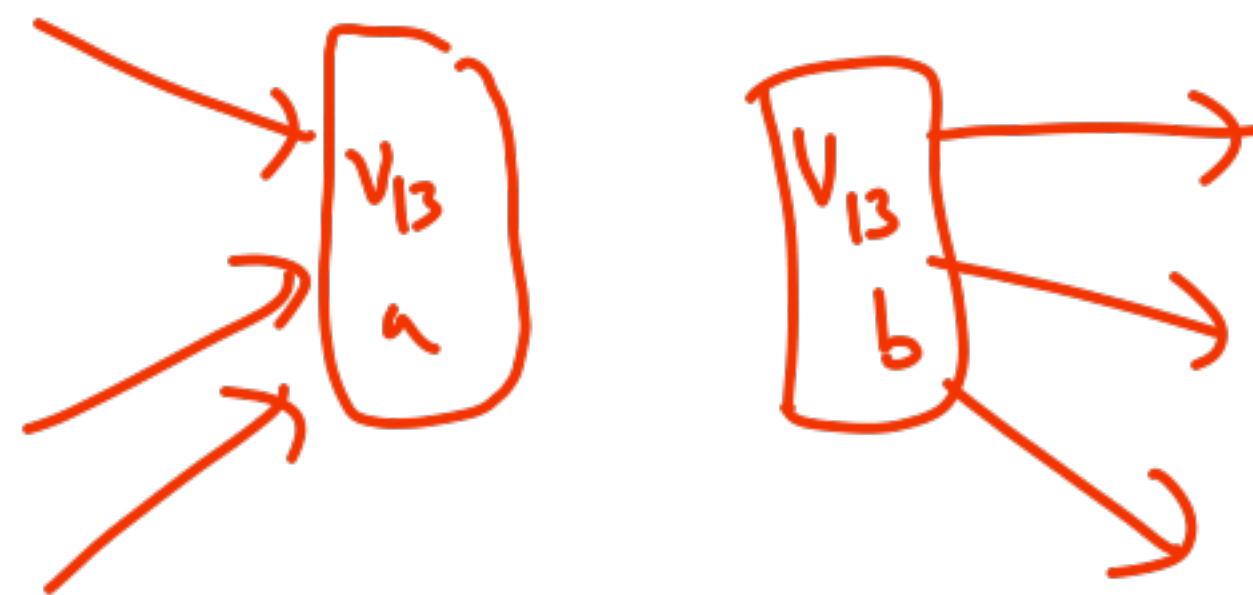
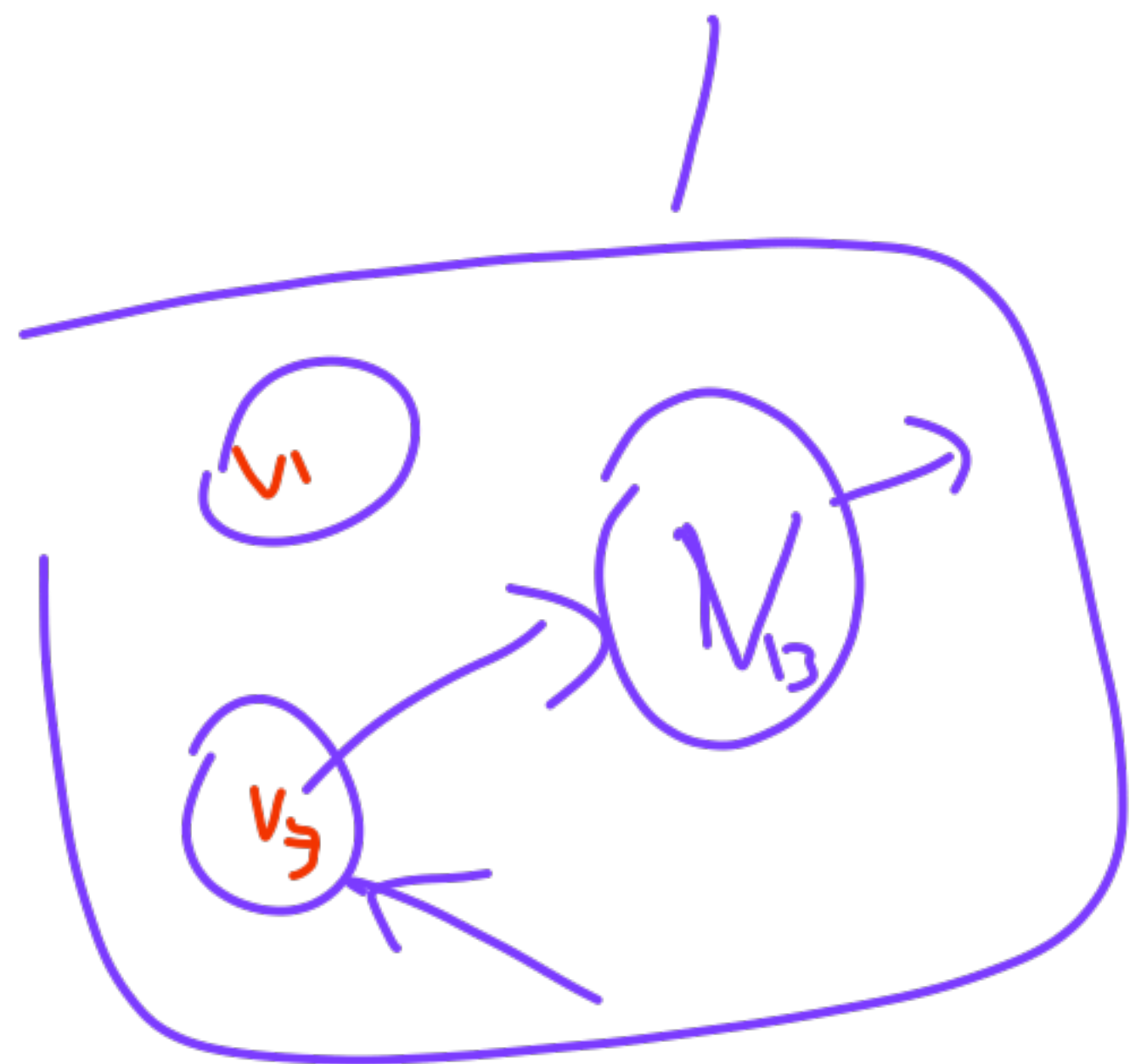
3 pairs of arrows
one for each literal

Soln-preserving.
(equi-satisfiable)

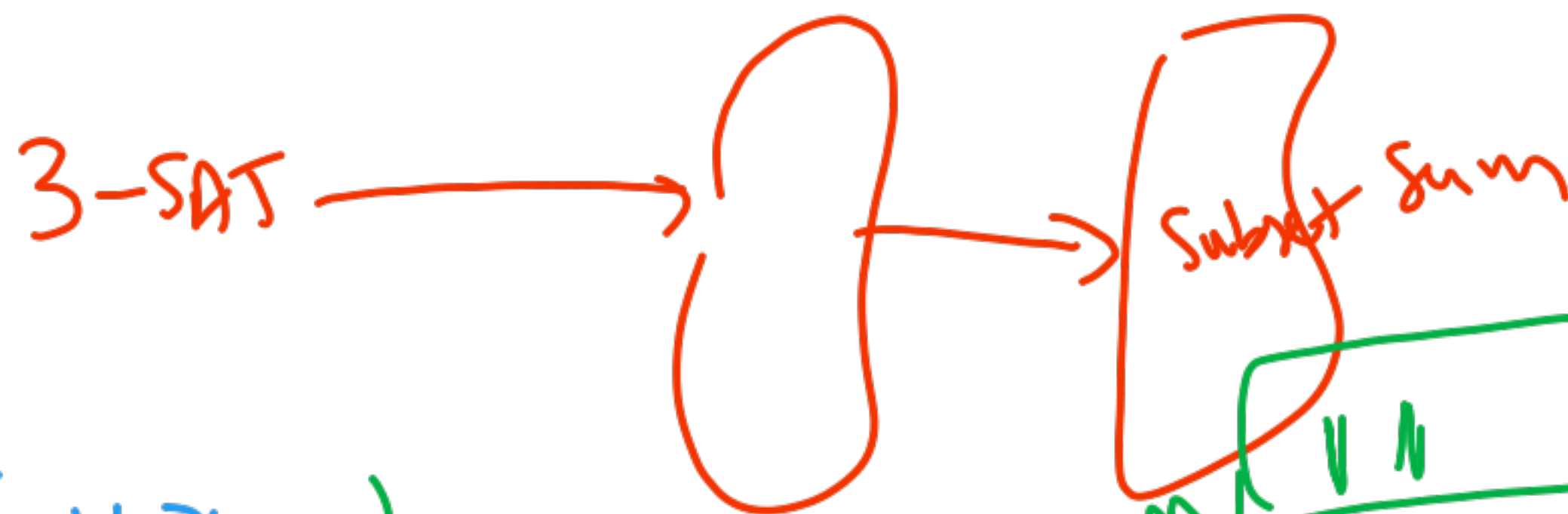
- C1. $\underline{x_1} \vee \bar{x_2} \vee x_4$
- C2. $\bar{x_1} \vee x_3 \vee \bar{x_4}$
- C3. $\bar{x_2} \vee \bar{x_3} \vee x_4$



3SAT $\rightarrow$ HC is NP-complete $\rightarrow$ LIP-path is NP-complete



Subset-Sum



$C_1 \quad x_1 \vee \bar{x}_2 \vee x_3$
 $C_2 \quad x_2 \vee \bar{x}_3 \vee \bar{x}_4$

$\left. \begin{array}{l} C_1 \\ C_2 \end{array} \right\} S \text{ has } 12 \text{ numbers}$

$S = \{n_1, \dots, n_k\}, M$

Subset $Q \subseteq S \rightarrow$ sum of numbers in $Q = M$
slack

M

	x_1	x_2	x_3	x_4	C_1	C_2
p_1	1	0	0	0	1	0
n_1	1	0	0	0	0	0
p_2	0	1	0	0	0	1
n_2	0	1	0	0	1	0
p_3	0	0	1	0	0	0
n_3	0	0	1	0	0	0
p_4	0	0	0	1	0	0
n_4	0	0	0	1	0	0
C_1	0	0	0	0	1	0
C_2	0	0	0	0	0	1

M

